

Database

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Database Documentation

Production uses normalized PostgreSQL with Prisma ORM as the required application data-access layer. The SQL schema is in `06Code/migrations/001_initial_schema.sql` , seed data is in `06Code/seeders/001_seed.sql` , and the ORM schema is in `06Code/prisma/schema.prisma` .

The SQL schema and Prisma schema are kept aligned for the academic model, including the `level_promotion_evaluations.evaluated_by` user relation and `evaluated_at` timestamp.

Main Tables

Table	Purpose
<code>branches</code>	Academy branches: Norte, Matriz, Sur Guamani, Tumbaco, Conocoto.
<code>roles</code> , <code>permissions</code> , <code>role_permissions</code>	Internal role/permission catalog controlled by the app.
<code>users</code>	Google-linked application users with internal role reference.

Table	Purpose
<code>students</code>	Student academic profile, branch and level B1/B2.
<code>teachers</code>	Teacher profile, branch and hourly rate.
<code>dance_categories</code> , <code>dance_styles</code> , <code>teacher_styles</code>	Dance catalog and teacher-style assignments.
<code>class_groups</code> , <code>class_sessions</code>	Academic groups and scheduled sessions.
<code>student_attendance_records</code>	Student attendance by session with unique student/session rule.
<code>teacher_attendance_records</code>	Teacher check-in/check-out records for hours and payment.
<code>absence_justifications</code>	Student/teacher absence review workflow.
<code>scholarship_rules</code> , <code>scholarship_evaluations</code>	Scholarship threshold and director evaluation records.
<code>level_promotion_evaluations</code>	B1 to B2 evaluation evidence.
<code>enrollment_requests</code>	Public landing enrollment requests.
<code>sessions</code>	Server-side session hashes, expiration and revocation.
<code>audit_logs</code>	Traceability for administrative and academic actions.

Normalization Notes

- User identity is separated from student/teacher academic profiles.
- Roles and permissions are normalized instead of hardcoding permissions in user rows.
- Dance categories and styles avoid repeated text inside class groups.
- Attendance records reference class sessions and students/teachers.
- Scholarship and level promotion approvals store evaluation evidence separately from candidate calculation.

Integrity Rules

- `student_attendance_records` has `UNIQUE(student_id, class_session_id)`.
- Level values are constrained to `B1` and `B2`.
- Attendance status is constrained to `present` , `absent` , `justified` , `late`.

- Scholarship percentage is constrained to `25`, `50`, `75`, `100`.
- Session tokens are stored as hashes.

ORM Runtime Selection

- `NODE_ENV=test` uses in-memory repositories for fast automated tests.
- `DB_DRIVER=prisma` and `DATABASE_URL` enable Prisma ORM repositories in production/deployment.
- `DB_DRIVER=pg` is retained only as a legacy fallback for the previous raw PostgreSQL repository.

Prisma Commands

```
cd 06Code
npm run db:generate
npm run db:push
```